

LOCAL ALGEBRA AND ISOLATED SINGULARITIES

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ABSTRACT. This study note establishes the foundational algebraic and geometric machinery required to analyze isolated hypersurface singularities, serving as a critical prerequisite for understanding the Milnor fibers and Stein fillings discussed by Plamenevskaya and Starkson. We first formalize the notion of a complex space germ $(X, 0)$ and its associated local ring $\mathcal{O}_{X,0}$ realized as a quotient of the ring of convergent power series $\mathbb{C}\{x_1, \dots, x_n\}$. Using the Jacobian Criterion, we explicitly compute the singular locus of a variety defined by a system of polynomials to determine if a singularity is isolated. These algebraic techniques are then applied to a concrete model case: verifying the isolation of the classical A_1 singularity defined by the hypersurface $x^2 + y^2 + z^2 = 0$ in $\mathbb{C}^3$. This calculation provides the essential framework for the subsequent study of smoothing techniques, link topology, and exotic contact structures on boundaries of Stein domains.

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1. COMPLEX SPACE GERMS

Definition 1.1 (Pointed Complex Space). A *pointed complex space* is a pair (X, x) consisting of a complex space X and a point $x \in X$. A *morphism* $f : (X, x) \rightarrow (Y, y)$ of pointed complex spaces is a morphism $f : X \rightarrow Y$ of complex spaces such that $f(x) = y$.

From definition 1.1, we describe pointed complex spaces as *objects*. The morphisms, on the other hand, are described as equivalence classes of the morphisms of pointed complex spaces defined in some open neighborhood. Note that we are saying there is a subset of elements that are considered "equal" to one another under a specific rule ($f(x) = y$). If we consider two neighborhoods U and V of $x \in X$, and $f : (U, x) \rightarrow (V, y)$ and $g : (U, x) \rightarrow (V, y)$, then the morphisms f and g are equivalent if there is an other neighbourhood, W such that $W \subset U \cap V$. Additionally, both f and g are restricted to W . To formalise our definitions, we will refer to *objects* **complex space germs**, and *morphisms* as **(holomorphic) map germs**.

Definition 1.2 (Composition and Isomorphism of Holomorphic Map Germs). If $f : (X, x) \rightarrow (Y, y)$, resp. $g : (Y, y) \rightarrow (Z, z)$, are holomorphic map germs, then they are represented by morphisms $f : (U, x) \rightarrow (Y, y)$, resp. $g : (V, y) \rightarrow (Z, z)$, of pointed complex spaces, where U, V are open neighborhoods of x, y , respectively. Then the *composition* $g \circ f : (X, x) \rightarrow (Z, z)$ is the holomorphic

map germ represented by $g \circ (f|_{f^{-1}(V) \cap U})$. The map germ $f : (X, x) \rightarrow (Y, y)$ is an *isomorphism* if there exists a map germ $h : (Y, y) \rightarrow (X, x)$ such that $f \circ h = \text{id}_{(Y, y)}$ and $h \circ f = \text{id}_{(X, x)}$.

An isomorphism between two complex space germs is the topological and algebraic equivalent of saying two spaces look completely identical when you zoom in infinitely close to the base points x and y .

- **The Inverse Property:** The map germ $h : (Y, y) \rightarrow (X, x)$ acts as a perfect geometric inverse to f .
- **The Role of id:** The symbol id denotes the *identity morphism* (the map that leaves every point unchanged).
- **Domain Matching:** The two compositions yield identities on their respective home spaces:

$$(f \circ h) = \text{id}_{(Y, y)} \quad (\text{Composition maps points in } Y \text{ back to themselves})$$

$$(h \circ f) = \text{id}_{(X, x)} \quad (\text{Composition maps points in } X \text{ back to themselves})$$

Thus, f and h undo each other completely in a sufficiently small neighborhood of the points.

Definition 1.3 (Representatives and Subgerms). If $U \subset X$ is an open neighborhood of x , then the germs (U, x) and (X, x) are isomorphic via the inclusion map $U \hookrightarrow X$. We identify the complex space germ (U, x) with (X, x) and call U a *representative* of the germ (X, x) .

Similarly, if $f : (X, x) \rightarrow (Y, y)$ is a holomorphic map germ and if $U \subset X$, resp. $V \subset Y$, are representatives of (X, x) , resp. (Y, y) , such that $f(U) \subset V$, then we call $f : U \rightarrow V$ a *representative* of the map germ f . If $X \subset Y$ is a complex subspace and $x \in X$, then (X, x) is called a *subgerm* of (Y, x) .

To fully parse the geometric intuition behind Definition 1.47(3), we break down its three core pillars:

1. **Inclusion Maps and Injectivity:** The hooked arrow symbol $\hookrightarrow$ denotes an *inclusion map* $\iota : U \hookrightarrow X$ defined by $\iota(u) = u$ for all $u \in U$. By definition, this map is strictly *injective* (one-to-one), meaning:

$$\forall u_1, u_2 \in U, \quad \iota(u_1) = \iota(u_2) \implies u_1 = u_2$$

In topology, this guarantees that the local geometry of the neighborhood U is preserved perfectly without distortion or collapsing when viewed inside the larger space X .

2. **The Extension to Neighborhood Representatives:** Because germs isolate properties at an infinitesimal local level, the global boundaries of a space do not matter.

- If U is a neighborhood of $x \in X$, the germ (U, x) is isomorphic to (X, x) , allowing us to call the open set U a *representative* of the germ.
- We extend this logic to a holomorphic map germ $f : (X, x) \rightarrow (Y, y)$. By picking suitable representative neighborhoods $U \subset X$ and $V \subset Y$ such that the image $f(U)$ is entirely contained within V , the restricted mapping $f : U \rightarrow V$ serves as a concrete *representative of the map germ*.

3. **Visualizing Subgerms:** A *subgerm* $(X, x) \subset (Y, x)$ is simply a local subset configuration. If a complex subspace X (such as an algebraic curve) is embedded within a larger complex space Y (such as a surface), and both pass through the basepoint x , then zooming in infinitely close to x reveals X acting as a local subset of Y .

Definitions 1.1–1.3 tend to hold for sufficiently small neighbourhoods. For the following definition, we define a *sheaf* as a mathematical tool that tracks local data across a topological space and ensures it glues together consistently.

Definition 1.4 (Analytic Local Ring, Reduced Germs, and Dimensions). If (X, x) is a germ, represented by the complex space X with structure sheaf $\mathcal{O}_X$, then the stalk $\mathcal{O}_{X,x}$ is called the (*analytic*) *local ring* of the germ (X, x) and also denoted by $\mathcal{O}_{(X,x)}$.

We call (X, x) *reduced* if the local ring $\mathcal{O}_{X,x}$ is reduced. Then we also say that X is *reduced at* x . Moreover, we set

$$\dim(X, x) := \dim_x X, \quad \text{edim}(X, x) := \text{edim}_x X$$

To understand complex space germs, we look closely at the mathematical properties of the local ring $\mathcal{O}_{X,x}$. When we say a local ring is reduced, we mean that it contains no non-zero nilpotent elements. In pure algebra, a nilpotent element is a non-zero function that yields zero when raised to a positive integer power ($f^n = 0$). Geometrically, if a ring contains these elements, the space behaves as though it has a hidden, infinitesimal thickness or fuzzy structure rather than clear, sharp coordinates. By restricting our study to a reduced germ, we ensure that every algebraic function is completely and uniquely determined by the actual geometric values it takes on the points of our space, which matches our physical intuition of shapes.

The definition introduces two distinct measurements of size: the intrinsic dimension, written as $\dim(X, x)$, and the embedding dimension, written as $\text{edim}(X, x)$. The intrinsic dimension tracks the true, inner geometric dimensions of the space itself. For instance, a complex line has an intrinsic dimension of one, while a complex surface has an intrinsic dimension of two. On the other hand, the embedding dimension measures the size of the smallest smooth ambient space, like $\mathbb{C}^N$, required to safely hold a local patch of our space without tearing or forcing it to intersect itself.

While it is easy to assume these two values are always equal, their relationship actually provides a powerful test for detecting geometric flaws. A fundamental rule in complex geometry states that the intrinsic dimension can never exceed the embedding dimension. If these two numbers match perfectly at a point, the space is entirely smooth and regular there, meaning it can be flattened out nicely into standard coordinates. However, if the embedding dimension is strictly larger than the intrinsic dimension, it signals that the space is crumpled or pinched at that specific coordinate. This structural mismatch is the exact mathematical definition of a singularity.

Recall the following regarding ideals.

Definition 1.5 (Ideal of a Ring). Let R be a commutative ring. A subset $I \subset R$ is called an *ideal* if it forms an additive subgroup of R and is closed under multiplication by any element of R . Explicitly, I must contain the zero element, satisfy $a + b \in I$ for all $a, b \in I$, and satisfy $r \cdot a \in I$ for all $a \in I$ and $r \in R$.

To understand how algebraic geometry bridges equations with geometric shapes, we rely heavily on the structural properties of ideals. In a ring of functions, an ideal can be thought of as a collection of mathematical constraints. The defining trait of an ideal is its absorption property under multiplication: multiplying any element inside the ideal by any arbitrary function in the entire ring always yields an output that lands back inside the ideal.

Geometrically, this behavior allows us to isolate the exact locations where a set of equations vanishes. For instance, if an ideal is generated by a specific polynomial, the ideal captures all the algebraic behavior that occurs along the geometric surface defined by setting that polynomial equal to zero. When we construct a local ring for a hypersurface germ by taking a quotient ring, we are algebraically dividing out by this ideal. This process forces our functions to ignore the ambient

outer space and behave strictly according to the boundaries and shapes dictated by the ideal, which is essential for isolating and analyzing local singularities.

Definition 1.6 (Subgerms Defined by Ideals and Hypersurface Singularities). Let (X, x) be a complex space germ, and let $I \subset \mathcal{O}_{X, x}$ be an ideal. Let $(U, \mathcal{O}_U)$ be a representative of (X, x) and $f_1, \dots, f_s \in \mathcal{O}_U(U)$ such that I is generated by the germs of $f_1, \dots, f_s$ at x . The closed complex subspace of U defined by $\mathcal{I} = \sum_{i=1}^s f_i \mathcal{O}_U$ defines a closed (analytic) subgerm

$$(V(I), x) := (V(\mathcal{I}), x) \subset (U, x) = (X, x)$$

of (X, x) , called the *closed (analytic) subgerm defined by I* .

If $I = \langle f \rangle \subset \mathcal{O}_{\mathbb{C}^n, p}$ is a principal ideal with $f \neq 0$, then the germ

$$(V(f), p) := (V(I), p) \subset (\mathbb{C}^n, p)$$

is called a *hypersurface singularity*. Hypersurface singularities in $(\mathbb{C}^2, p)$ are called *plane curve singularities*.

To understand how a subgerm is constructed, we trace the mathematical information flowing between our previously defined concepts. We begin with a global complex space, but because we only care about local behaviors, we look entirely within the stalk $\mathcal{O}_{X, x}$, which is the analytic local ring containing functions zoomed infinitely close to the point x . Within this ring, we isolate an ideal I , which acts as our algebraic constraint. To calculate with this ideal, we choose a local open neighborhood U to act as a representative of our germ space. On this neighborhood, the structure sheaf $\mathcal{O}_U$ provides us with actual, touchable holomorphic functions $f_1, \dots, f_s$ that serve as the generators of our ideal.

The notation $V(I)$ denotes the vanishing locus, which is the exact geometric collection of points where all of our generating functions equal zero simultaneously. By gathering these coordinates within our neighborhood, we construct a well-defined closed complex subspace. When we look at this vanishing set near our basepoint, it forms a subgerm $(V(I), x)$ nestled inside our original ambient germ (X, x) . This entire setup is simply the mathematically rigorous way of stating that we are isolating a local geometric shape defined by a specific system of equations passing through a point.

The definition narrows down to a vital special case when our ambient space is standard smooth complex space $(\mathbb{C}^n, p)$ and our ideal is principal, meaning it is generated by a single non-zero polynomial f . This produces a hypersurface singularity. If this single equation contains a sharp or pinched point where all of its partial derivatives vanish simultaneously, it violates regular smooth behavior. Because it forms a geometry where the embedding dimension is strictly greater than the intrinsic dimension, it provides the standard model for studying isolated singular points.

This exact algebraic mechanism serves as the literal launchpad for the contact geometry discussed in the Plamenevskaya–Starkson paper on unexpected Stein fillings. When you construct a hypersurface singularity $(V(f), 0)$ centered at the origin in $\mathbb{C}^n$, you can intersect this pinched geometric shape with an infinitesimally small, smooth odd-dimensional sphere S^{2n-1} . The boundary of this intersection forms a smooth manifold known as the link of the singularity. This link naturally inherits a canonical, tight contact structure dictated by the complex structure of the surrounding ambient space.

In standard symplectic topology, the Milnor fiber—which is the smooth manifold obtained by perturbing the singular equation slightly from $f = 0$ to $f = \epsilon$ —provides a natural geometric filling for this contact boundary known as a Stein filling. The core breakthrough of the Plamenevskaya–Starkson paper relies on looking at these exact types of hypersurface singularity models to build

exotic, completely unexpected Stein fillings that do not originate from standard Milnor geometric smoothings.

We define the fiber $(f^{-1}(y), x)$ of a morphism $f : (X, x) \rightarrow (Y, y)$ of complex space germs to be the germ of the inverse image of the target point. This fiber is represented algebraically by the quotient of the local ring $\mathcal{O}_{X,x}/\mathfrak{m}_{Y,y}\mathcal{O}_{X,x}$.

2. SINGULAR LOCUS AND DIFFERENTIAL FORMS

Definition 2.1 (Regular Points, Singular Points, and the Singular Locus). A point $x \in X$ is a *regular* (or *smooth*) point of X if and only if the local ring $\mathcal{O}_{X,x}$ is a regular local ring. Conversely, x is a *singular point* if and only if it is not regular. The set of all singular points of X is referred to as the *singular locus* of X and is denoted by $\text{Sing}(X)$.

The singular locus, denoted $\text{Sing}(X)$, is the set of points where the variety is not "well-behaved" or "smooth". That is essentially what this definition is saying. From a geometric perspective, these are the locations where the space pinches, creases, or intersects itself, meaning you cannot approximate the region smoothly. Algebraically, this means the local ring fails to be regular, indicating that the minimal number of algebraic generators needed to define the functions near that point exceeds the intrinsic dimension of the space.

Theorem 2.2 (The Jacobian Criterion for Complex Analytic Germs). *Let $X \subset \mathbb{C}^n$ be a complex analytic space defined locally near the origin $0 \in \mathbb{C}^n$ by a principal or radical ideal $I = \langle f_1, \dots, f_m \rangle \subset \mathbb{O}_{\mathbb{C}^n,0}$, where each f_i is a holomorphic function germ. Let $d = \dim(X, 0)$ be the intrinsic dimension of the variety near 0.*

Define the local Jacobian matrix $J(f)$ of partial derivatives evaluated at a point $p \in X$ by:

$$J(f)(p) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(p) & \cdots & \frac{\partial f_1}{\partial x_n}(p) \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1}(p) & \cdots & \frac{\partial f_m}{\partial x_n}(p) \end{pmatrix}$$

Then, the point p is a regular (smooth) point of X if and only if the rank of the Jacobian matrix satisfies:

$$\text{rank}(J(f)(p)) = n - d = \text{codim}(X, p)$$

Conversely, $p \in \text{Sing}(X)$ is a singular point of the variety if and only if the rank drops below this critical codimension value:

$$\text{rank}(J(f)(p)) < n - d$$

The Jacobian criterion converts our geometric definitions of smoothness and singularity into an explicit rank problem in linear algebra. The expression $n - d$ represents the codimension of the variety, which counts the total number of independent functional equations theoretically required to define our space inside the ambient space $\mathbb{C}^n$. When the rank of the Jacobian matrix equals this exact value, it mathematically confirms that the linear approximations of our equations are completely independent. This independence allows us to apply the Implicit Function Theorem, guaranteeing that we can locally represent our variety as a perfectly smooth complex manifold.

If the rank drops below the codimension value, it reveals that the partial derivatives have collapsed or vanished simultaneously, indicating that our defining equations fail to provide a linearly independent frame at that specific location. Geometrically, this rank drop is the telltale sign that the space contains a pinching point or intersection crease. Because it is computationally straightforward

to test where the maximal determinants (minors) of a matrix drop to zero, the Jacobian criterion is the primary tool used to explicitly compute the boundaries and constraints of the singular locus $\text{Sing}(X)$.

3. TASK: THE A_1 SINGULARITY

Given the hypersurface X in $\mathbb{C}^3$ defined by the equation:

$$f(x, y, z) = x^2 + y^2 + z^2 = 0$$

1. **Define the Local Ring:** Write the expression for $\mathcal{O}_{X,0}$ as a quotient of the ring of convergent power series $\mathbb{C}\{x, y, z\}$.
2. **Compute the Gradient:** Calculate the partial derivatives $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$, and $\frac{\partial f}{\partial z}$.
3. **Identify the Singular Locus:** Determine the set of points where all partial derivatives vanish simultaneously.

Solutions to the A_1 Singularity

1. *Local Ring Formulation.* The local ring $\mathcal{O}_{X,0}$ of the hypersurface germ $(X, 0)$ is constructed by taking the quotient of the ambient ring of convergent power series by the principal ideal generated by the defining polynomial $f(x, y, z)$. Thus, the analytic local ring is given by:

$$\mathcal{O}_{X,0} = \mathbb{C}\{x, y, z\} / \langle x^2 + y^2 + z^2 \rangle$$

where $\langle x^2 + y^2 + z^2 \rangle = \{g \cdot (x^2 + y^2 + z^2) \mid g \in \mathbb{C}\{x, y, z\}\}$ represents the ideal of geometric relations holding the variety to the surface.

2. *Gradient Computation.* To find the local linear approximations of the hypersurface, we compute the gradient vector ∇f by taking the partial derivatives with respect to each independent ambient coordinate variable:

$$\frac{\partial f}{\partial x} = 2x, \quad \frac{\partial f}{\partial y} = 2y, \quad \frac{\partial f}{\partial z} = 2z$$

This yields the Jacobian row matrix $J(f) = \begin{pmatrix} 2x & 2y & 2z \end{pmatrix}$.

3. *Identification and Isolation of the Singular Locus.* According to the Jacobian criterion, the singular locus $\text{Sing}(X)$ consists of all coordinates where the rank of the Jacobian matrix drops strictly below the codimension of the variety ($n - d = 3 - 2 = 1$).

Evaluating the matrix at the origin by plugging in $(0, 0, 0)$ yields:

$$J(f)(0, 0, 0) = \begin{pmatrix} 0 & 0 & 0 \end{pmatrix}$$

Since all matrix entries collapse to zero, $\text{rank}(J(f)(0, 0, 0)) = 0$. Because $0 < 1$, the origin is a singular point.

Furthermore, setting the gradient components simultaneously to zero forms the system:

$$2x = 0, \quad 2y = 0, \quad 2z = 0$$

This system has the unique solution $(x, y, z) = (0, 0, 0)$. Consequently, the singular locus contains exactly one point:

$$\text{Sing}(X) = \{(0, 0, 0)\}$$

Because there are no other points satisfying the vanishing condition in any neighborhood of the origin, the A_1 hypersurface singularity is strictly verified to be an *isolated singularity*.

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